

DBMS PROJECT

Dharohar Mehta - 2024186

Shivansh Singla - 2024532

Rohan - 2024475

Project Group - 48

Tut Group - 4

TA - Raghav Verma and Rahul Pardasani

Task 1: Project Scope

Pharma Wholesaler Inventory Management System

1. Introduction

The pharmaceutical supply chain requires strict inventory control, accurate records, regulatory compliance, and timely order fulfillment. Pharmaceutical wholesalers serve as key intermediaries between manufacturers and pharmacies or hospitals. Without proper systems, issues such as stockouts, expired drugs, pricing errors, and compliance failures can occur.

This project focuses on designing a database-driven Pharma Wholesaler Inventory Management System, emphasizing backend database design, constraints, transactions, triggers, and SQL operations, supported by a simple demonstration interface.

2. Business Problem Statement

A pharmaceutical wholesaler must:

- Track thousands of medicines, including batch numbers and expiry dates.
- Maintain real-time inventory levels.
- Process purchase orders from manufacturers
- Fulfill sales orders from retailers.
- Ensure that expired or recalled drugs are not sold.
- Maintain transaction consistency during concurrent operations.

Without a robust database system, the wholesaler may face issues such as:

- Sale of expired medications
- Incorrect stock levels
- Revenue losses
- Regulatory non-compliance

3. Project Objectives

The objectives of this project are as follows:

- To design a normalized relational database tailored for a pharmaceutical wholesaler.
- To enforce data integrity and business rules through constraints and triggers.
- To support inventory, supplier, customer, and order management functionalities.
- To implement complex SQL queries for analytics and reporting purposes.

- To demonstrate database transactions, including scenarios involving conflicting transactions.
- To integrate the database with a front-end user interface for real-world simulation.

4. Scope of the System

In Scope:

- Management of medicine master data
- Batch-wise inventory tracking (including expiry and pricing)
- Supplier (manufacturer) management
- Customer (retailer/pharmacy) management
- Processing of purchase and sales orders
- Real-time inventory updates via transactions
- Automated checks utilizing database triggers
- Reporting and analytics conducted through SQL queries

Out of Scope:

- Integration of online payment gateways
- Real-time government drug regulation APIs
- Logistics and delivery route optimization

5. Major Functional Requirements

- **Medicine Management:** Add, update, and view details of medicines while maintaining category, dosage, and pricing information.
- **Inventory Management:** Track quantities per batch, store expiry dates, and prevent the sale of expired drugs.
- **Supplier Management:** Store supplier details and track purchase history.
- **Customer Management:** Manage details of retailers and hospitals, while maintaining order history.
- **Order Processing:** Process purchase orders (stock in) and sales orders (stock out) with automatic inventory updates.
- **Triggers & Constraints:** Block the insertion of expired batches, automatically update inventory upon sales, and prevent harmful stock levels.
- **Transactions:** Handle concurrent order placements, facilitate rollbacks on failures, and demonstrate conflicting transactions.

6. Non-Functional Requirements

- **Data Integrity:** Ensured through primary keys (PK), foreign keys (FK), CHECK, and UNIQUE constraints.
- **Consistency:** Transactions adhere to ACID principles.
- **Scalability:** Designed to accommodate large datasets in the medical field.
- **Usability:** Incorporates a simple user interface for demonstration.
- **Reliability:** Features error handling achieved via triggers and transactions.

7. Target Users

- **Inventory Manager:** Monitors stock levels and expiry information.
- **Sales Executive:** Processes customer orders.
- **Admin:** Oversees suppliers, medicines, and report generation.

8. **Database-Centric Design Approach**

This project emphasizes:

- Robust conceptual and relational modeling
- Extensive use of SQL
- Enforcement of business logic at the database level
- Minimal yet effective application-layer logic